

Product Submission to the Therapeutic Advisory Service
Meeting Date: 8th September 2015

Toujeo: U300 Insulin Glargine requested by Dr Kong, Consultant Diabetologist

TAS Minutes / Decision:

This is a licensed, more concentrated product which could be used in place of U500 insulin. TAS members were broadly supportive of use but concerns expressed about the number of insulins available, particularly as biosimilar insulin glargine has just been launched. The speciality is asked to review current choices with a view to rationalisation. Further members wished to know if this would replace U500 insulin completely

TAS decision: Deferred

October 15:

Feedback from diabetologists was that this couldn't directly replace an existing product but it has a specific place in therapy. Re rationalisation of products there were some concerns within the speciality that not having an extensive range of products available could result in missed doses with dire consequences. Notwithstanding they would look at currently available products with a view to rationalisation but with the caveat above. LG advised that she could supply current primary care data to support this review. There were still governance and risk concerns expressed by the committee about the number of products available. It was agreed by TAS that following establishing current product use if we could identify small usage of specific products then controlled switching could be considered.

TAS decision: Supported for use. Review of current prescribing to be undertaken with a view to reducing the number of products available

INFORMATION SUPPLIED BY REQUESTOR

Consultant	Dr Marie-France Kong	Clinical Management Group	Acute Medicine / ED Specialist Medicine
Speciality	Diabetes	CMG Support for Funding	Pending
Drug Name Generic	Insulin Glargine 300 units/mL	Drug Name Proprietary	Toujeo®
Formulation and strength		Insulin glargine 300 units/mL	
Typical Maintenance dose		30 to 450 units daily	
Clinical Indication: Diabetes mellitus in adults			
Licensed Indication? Yes		Licensed product? Yes	
Are patients currently using this agent in UHL as part of a clinical trial? No		How many patients per year? Initially 50-60 per year	

INFORMATION SUPPLIED BY MEDICINES INFORMATION

SUMMARY

- Toujeo® (Insulin glargine 300units/mL) is a long acting insulin analogue for treatment of diabetes mellitus in adults
- Several randomised controlled trials have shown Toujeo® to be non-inferior to Lantus® (Insulin glargine 100units/mL) in terms of safety and efficacy
- Toujeo® was associated with a lower incidence of confirmed hypoglycaemia or nocturnal hypoglycaemic episodes when compared to Lantus®

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The information in this review is believed to be true and accurate. It is issued on the understanding that it is the best available from the resources at our disposal at the time of issue.

- The cost is similar to other long acting insulin analogues and lower than the imported, unlicensed U500 insulin

COST COMPARISON * For 28 days treatment	Product	Manufacturer	Dose	Cost per dose	Cost for whole packs dispensed**
	Toujeo® (Insulin glargine U300)	Sanofi-Aventis	50 units daily	£1.23	£44.17 (4x1.5mL pens providing 1800 units)
	Lantus® (Insulin glargine U100)	Sanofi-Aventis	50 units daily	£1.38	£41.50 (5x3mL pens providing 1500 units)
	Levemir® (Insulin detemir U100)	Novo-Nordisk	50 units daily	£1.40	£42.00 (5x3mL pens providing 1500 units)
	Tresiba® (Insulin degludec U200)	Novo-Nordisk	50 units daily	£2.40	£86.40 (3x3mL pens providing 1800 units)
	U500 Insulin (unlicensed)	Lilly USA (imported)	50 units daily	£1.25	£250.74 (1 20mL vial providing 10000units)

* eMIMS 02/09/2015

**Hospital prices may vary

Patent Expiries of note: Insulin glargine (brand name not specified): May 2015

Current Status

TAS	Not previously reviewed
NICE	ESNM due October 2015
SMC	Due September 2015
Pharmaceutical Risk assessment	Low

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Literature Search

Background

In patients with type 2 diabetes requiring insulin, basal insulin is often preferred as they are associated with fewer incidents of hypoglycaemia and less weight gain. This new 300unit/ml insulin glargine has been shown to be safe and effective when compared to 100units/ml insulin glargine [1].

Current Use (Local)

Long acting insulins on the Leicestershire Medicines Formulary include insulin glargine, insulin detemir and insulin degludec. In addition there are several patients managed on U500 insulin (Humulin R) [2].

Current Use (Other centres)

Not known

Clinical Trials

The key trials investigating the safety and efficacy of Toujeo® come from the EDITION series. EDITION 1, 2 and 3 were all multi-centre, randomised, open-label, two arm, parallel group, phase III studies enrolling adults with a diagnosis of type 2 diabetes mellitus. Each study included a 2-week screening phase, a 6-month treatment period followed by a 6-month safety extension period. EDITION 4 compared insulin glargine 300 units/mL with insulin glargine 100 units/mL in patients with type 1 diabetes.

EDITION 1

Patients enrolled in EDITION 1 were current users of basal (≥ 42 units daily) and mealtime insulin therapy with an HbA1c between 7.0 and 10%. Of the 807 patients enrolled, 404 received insulin glargine 300 units/mL (Gla-300) and 403 received insulin glargine 100 units/mL (Gla-100). The primary end point was HbA1c change from baseline to month 6. The main secondary efficacy endpoint was the percentage of patients with one or more confirmed (≤ 3.9 mmol/L) or severe nocturnal hypoglycaemic event reported between week 9 and month 6. Baseline HbA1c was 8.15% and 8.16% for the Gla-300 and Gla-100 groups respectively. At the end of the 6 month treatment period mean HbA1c had decreased to 7.25% (SD 0.85) with Gla-300 vs. 7.28% (SD 0.92) with Gla-100 which was shown to meet the non-inferiority criteria (LS mean change -0.83% (SE 0.06) for both groups, difference -0.00%; 95%CI -0.11 to 0.11). The proportion of participants with one or more confirmed hypoglycaemic event or severe nocturnal hypoglycaemia in the Gla-300 group was 36% compared to 46% in the Gla-100 group ($p=0.0045$) [3].

EDITION 2

Patients enrolled in EDITION 2 were current users of basal insulin (≥ 42 units daily) and oral antihyperglycaemic drugs (except sulphonylureas) with an HbA1c between 7.0 and 10%. A total of 811 patients were randomised to receive Gla-300 ($n=404$) or Gla-100 ($n=407$). Efficacy and safety analysis included change in HbA1c, total daily insulin dose, body weight and confirmed (≤ 3.9 mmol/L) hypoglycaemic events. The mean HbA1c decreased from baseline to month 12 in both groups. At month 12 the mean HbA1c for Gla-300 was 7.62% (SD 1.03) vs. 7.64% (SD 1.21) in the Gla-100 group (LS mean difference -0.06; 95%CI: -0.22 to 0.1%; p value not stated). Over the treatment period insulin doses increased in both groups. The mean dose of Gla-300 was 0.97 units/kg/day compared to 0.87 units/kg/day of Gla-100 (mean difference 0.11 (SE 0.02), $p<0.0001$ at month-12). The mean change (SD) in body weight was statistically significantly lower with Gla-300 (0.4 (4.1) kg) than Gla-100 (1.2 (3.6) kg) (LS mean difference at month 12 was -0.7; 95% CI: -1.3 to -0.2kg; $p=0.009$). The number of patients with confirmed hypoglycaemic events was lower in the Gla-300 group compared to the Gla-100 group (numbers not stated) however this was not statistically significant [4].

EDITION 3

Patients enrolled EDITION 3 were insulin naive patients using oral anti hyperglycaemics, with an HbA1C between 7 and 11%. Patients were randomised to receive Gla-300 ($n=439$) or Gla-100 ($n=439$). The primary endpoint was change in HbA1C from baseline to month 6. The main secondary end point was the percentage of patients experiencing one or more nocturnal confirmed or severe hypoglycaemic event reported between week 9 and month 6. Additional secondary endpoints included mean change in basal insulin dose, and body weight. The mean decrease in HbA1c was equivalent in the two treatment groups at month 6 (LS mean difference in change of HbA1c: 0.04%; 95% CI -0.09 to 0.17). Therefore non-inferiority was shown. The percentage of patients experiencing at least one nocturnal hypoglycaemic event was 16% with Gla-300 vs. 17% with Gla-100 (RR 0.89; 95% CI 0.66 to 1.2). The mean insulin dose increased in both groups over the treatment period but more so in the Gla-300 group to mean dose of 0.62 (0.29) units/kg/day with Gla-300 and to 0.53 (0.24) units/kg/day with Gla-100. Weight gain was lower with Gla-300 (LS mean increase 0.49kg; 95% CI 0.14 to 0.83) compared to Gla-100 (LS mean increase 0.71kg 95% CI 0.36 to 1.06) although this difference was found to be non-significant [5].

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EDITION 4

Patients enrolled in this study had type 1 diabetes, an average HbA1c of 8.1% and used basal-bolus insulin. A total of 549 patients were randomised 1:1:1:1 to received once daily Gla-300 or Gla-100, morning or evening, while continuing their mealtime insulin. Gla-300 was non-inferior to Gla-100 for HbA1c change from baseline (primary endpoint) (LS mean change [SE] - 0.40 [0.05] % and - 0.44 [0.05] %; LS mean difference 0.04 [95% CI: - 0.10 to 0.19] %). Event rates of confirmed (≤ 3.9 mmol/L) or severe hypoglycemia at any time (24 h) were similar for the two groups; nocturnal hypoglycemia was lower for Gla-300 versus Gla-100 during the first 8 weeks. Severe hypoglycemia was observed in 6.6% (Gla-300) and 9.5% (Gla-100) of participants. Neither glycemic control nor hypoglycemia differed between insulins or times for morning and evening injection groups. Total insulin dose increased more with Gla-300 versus Gla-100 (change from baseline + 0.19 versus + 0.10 U/kg). However weight gain was significantly lower with Gla-300 versus Gla-100 (LS mean difference - 0.56 [-1.09 to -0.03] kg, $p = 0.037$) [7].

Adverse Effects / safety issues

Across the EDITION 1, 2 and 3 studies no between-treatment differences in safety profile were identified. Treatment-emergent adverse events (TEAEs) were reported by 712 (57.3%) participants in the Gla-300 groups and 669 (53.7%) participants in the Gla-100 groups. The most common TEAEs were infection and infestations, nervous system disorders, GI disorders, musculoskeletal and connective tissue disorders and administration site conditions. In total, across all 3 studies 17 (1.4%) Gla-300 participants and 16 (1.3%) Gla-100 participants discontinued treatment due to a TEAE. No deaths were considered related to the study medication in any of the trials [1]. No difference was seen between the groups in the EDITION 4 study [6]

Other points to note

- NICE are considering an insulin glargine biosimilar and are due to publish an ESNM for this in October 2015.
- The U100 and U300 glargine preparations are not bio equivalent and a dose conversion is needed. When switching from insulin glargine 100 units/ml to Toujeo a higher dose (approximately 10-18%) may be needed to achieve target ranges for plasma glucose levels [7]
- U500 is excluded to tariff; Insulin glargine 300units/mL is not.

References

1. Ritzel R., Roussel R., Bolli G.B., et al. Patient-level meta-analysis of the EDITION 1, 2 and 3 studies: glycemic control and hypoglycaemia with new insulin glargine 300 U/ml versus glargine 100 U/ml in people with type 2 diabetes. *Diabetes Obesity and Metabolism*. 2015; 17:859-867.
2. Leicestershire Medicines Formulary available at <http://netformulary.xuhl-tr.nhs.uk/netformulary/> <accessed <01/09/2015>
3. Riddle M.C., Bolli G.B., Ziemer M., et al. New Insulin Glargine 300 Units/mL versus Glargine 100 Units/mL in People With Type 2 Diabetes Using Basal and Mealtime Insulin: Glucose Control and Hypoglycemia in a 6-month Randomised Controlled Trial (EDITION 1). *Diabetes Care*. 2014; 37: 2755-2762
4. Yki-Jarvinen H., Bergenstal R.M., Bolli G.B., et al. Glycaemic control and hypoglycaemia with new insulin glargine 300U/mL versus glargine 100U/mL in people with type 2 diabetes using basal insulin and oral antihyperglycaemic drugs (EDITION 2 randomised 12-month trial including 6-month extension) *Diabetes Care*. 2014; 37(12):3235-3243
5. Bolli G.B., Riddle M.C., Bergenstal R.M., et al. New insulin glargine 300 U/ml compared with glargine 100 U/ml in insulin naive people with type 2 diabetes on oral glucose-lowering drugs: a randomised controlled trial (EDITION 3). *Diabetes, Obesity and Metabolism*. 2015; 17: 386-394
6. Home P.D., Bergenstal R., Riddle M.C., et al. Glycemic control and hypoglycaemia with new insulin glargine 300 U/mL in people with type 1 diabetes (EDITION 4). *Diabetes Technology and Therapeutics*. 2015; 17:A103-A104
7. SPC Toujeo[®] (Insulin glargine) accessed at www.medicines.org.uk last updated 22/06/2015 <accessed 01/09/2015>

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Leicestershire Medicines Information Service
Date 02/09/2015

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Additional Information supplied by requestor

Place in therapy / Advantages over existing treatments. Include which drugs (if any) this will replace

Toujeo® is a long acting analogue insulin with a duration of action >24 hours after a single injection and it shows low day-to-day variability of glucose-lowering effect within-subjects. Compared to Lantus, Toujeo® offers a more stable and prolonged activity profile, lasting beyond 24 hours, similar rates of confirmed (≤ 3.9 mmol/L) total hypoglycaemia in patients with type 1 diabetes, with similar HbA1c reduction and a lower incidence of confirmed (≤ 3.9 mmol/L) total hypoglycaemia in patients with type 2 diabetes, with similar HbA1c reduction and once daily dosing with the benefit of flexibility when needed (± 3 hours). It is a new basal higher strength insulin preparation available in the UK for both type 1 and type 2 patients requiring analogue long acting insulin. In addition it will benefit patients who are insulin resistant and requiring large doses of insulin. It is administered with a pre-filled pen injector that is calibrated for the strength of insulin therefore significantly reducing the risk associated with U500 Humulin R®. (Patients on U500 currently use U100 insulin syringes which administer 5x the stated number of units *on the syringe* per injection.)

*Shelf life of 30 months vs. 28 days. Potential for less wastage according to dose required. Toujeo® will also be an alternative to unlicensed U500 Humulin R® in treating insulin resistant patients who require >200 units insulin per day to achieve adequate glycaemic control. This includes switching a proportion of those currently taking doses of U500 Humulin R® at the lower end of the dose range, and thereafter starting insulin resistant patients on this product before progressing to unlicensed U500 Humulin R®. Toujeo® (U300 insulin glargine) would be used to provide the basal requirement with prandial doses of standard (U100) strength short-acting insulin. The cost of Toujeo® is comparable to insulin glargine U100. Procurement, prescribing, and dispensing are standard rather than a unique process requiring additional pharmacy resources to administer for an unlicensed drug.

*Not able to confirm 28 day shelf life, expiry is not reduced by out patient pharmacy and products have standard shelf life when they are received into stock.

DOI: Members of the Department have at various times received speaker fees, sponsored places at meetings, and educational support from Sanofi. Individual consultants serve or have served on its advisory board.

References provided:

1. Riddle M et al. on behalf of the EDITION 1 Study Investigators. New Insulin Glargine 300 Units/mL Versus Glargine 100 Units/mL in People with Type 2 Diabetes Using Basal and Mealtime Insulin: Glucose Control and Hypoglycemia in a 6-Month Randomized Controlled Trial (EDITION 1). *Diabetes Care* 2014; 37:2755–2762 | DOI: 10.2337/dc14-0991.
2. Yki-Jarvinen H et al on behalf of the EDITION 2 Study Investigators. New Insulin Glargine 300Units/mL Versus Glargine 100 Units/mL in People With Type 2 Diabetes Using Oral Agents and Basal Insulin: Glucose Control and Hypoglycemia in a 6-Month Randomized Controlled Trial (EDITION 2). *Diabetes Care* 2014; 37:3235–3243 | DOI: 10.2337/dc14-0990.

Pharmacoeconomic category

D – New Treatment less expensive or similar cost and more effective than comparable therapy.

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